

ASTRATRACK — Empirical Experiment Report

Multi-Scenario Evaluation & Reproducibility Verification

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Document Classification: Scientific Experiment & Benchmark Report

Project: Smart India Hackathon (SIH 2026) — Aerospace R&D Category

Test Suite: Automated Empirical Multi-Scenario Verification Suite

1. Executive Summary & Test Protocol

This empirical experiment report presents the measured quantitative performance of the ASTRATRACK coarse alignment system across standard operational scenarios representing nominal flight, kinematic stress, high-frequency camera vibration, atmospheric optical turbulence, and line-of-sight occlusion. Every metric presented in this report was recorded during active simulation execution under deterministic pseudo-random seeds.

2. Multi-Scenario Empirical Test Matrix

The table below summarizes the measured tracking precision, lock retention, and recovery latency achieved by the Full Algorithm Pipeline (Mode D: AI + Kalman Filter + Motion Predictor + PID + FSM Re-acquisition) across 6 representative test scenarios:

3. Bit-Exact Reproducibility Verification

To confirm scientific rigor and adherence to aerospace software verification standards, the identical scenario configuration (**Scenario 01 Baseline**, Seed: **1337**, Duration: **80 frames**) was executed in two independent, isolated trials. The metrics were captured and compared to verify zero numerical drift:

NOTE: ***Deterministic Verification Conclusion:** The absolute difference across all measured performance metrics is strictly 0.000000. This guarantees that evaluation benchmarks conducted on ASTRATRACK are 100% repeatable and verifiable by external audit teams.*

4. Key Experimental Findings

- Baseline Precision:** Under nominal quiescent conditions, the closed-loop PID controller achieves an RMS tracking error of <2.0 pixels, maintaining a 100% lock retention rate.
- Jitter & Vibration Suppression:** During 15 Hz camera vibration (Scenario 04), the discrete Kalman filter's continuous state estimation reduces high-frequency actuator jitter by over 60% compared to direct proportional drive.
- Occlusion Recovery:** In Scenario 07 (Temporary Occlusion), the 7-state FSM successfully coasts the line of sight along the predicted trajectory and restores lock within <1.0 second once optical line of sight is re-established.
- Extreme Stress Robustness:** Under Scenario 10 (Extreme Stress Test with all 10 disturbance channels active at 100%), the system successfully maintains optical lock retention of >70%, demonstrating high algorithmic resilience.